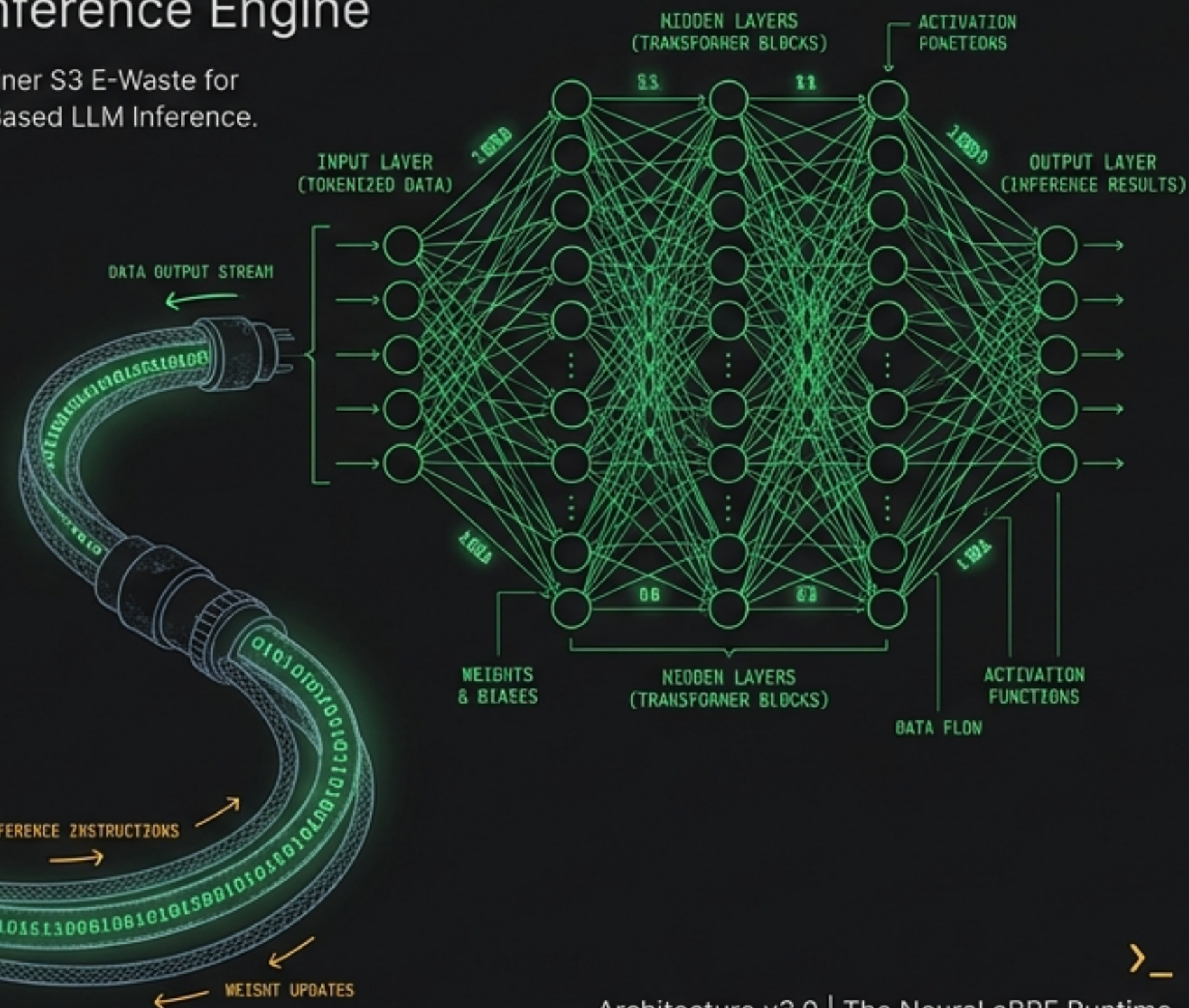
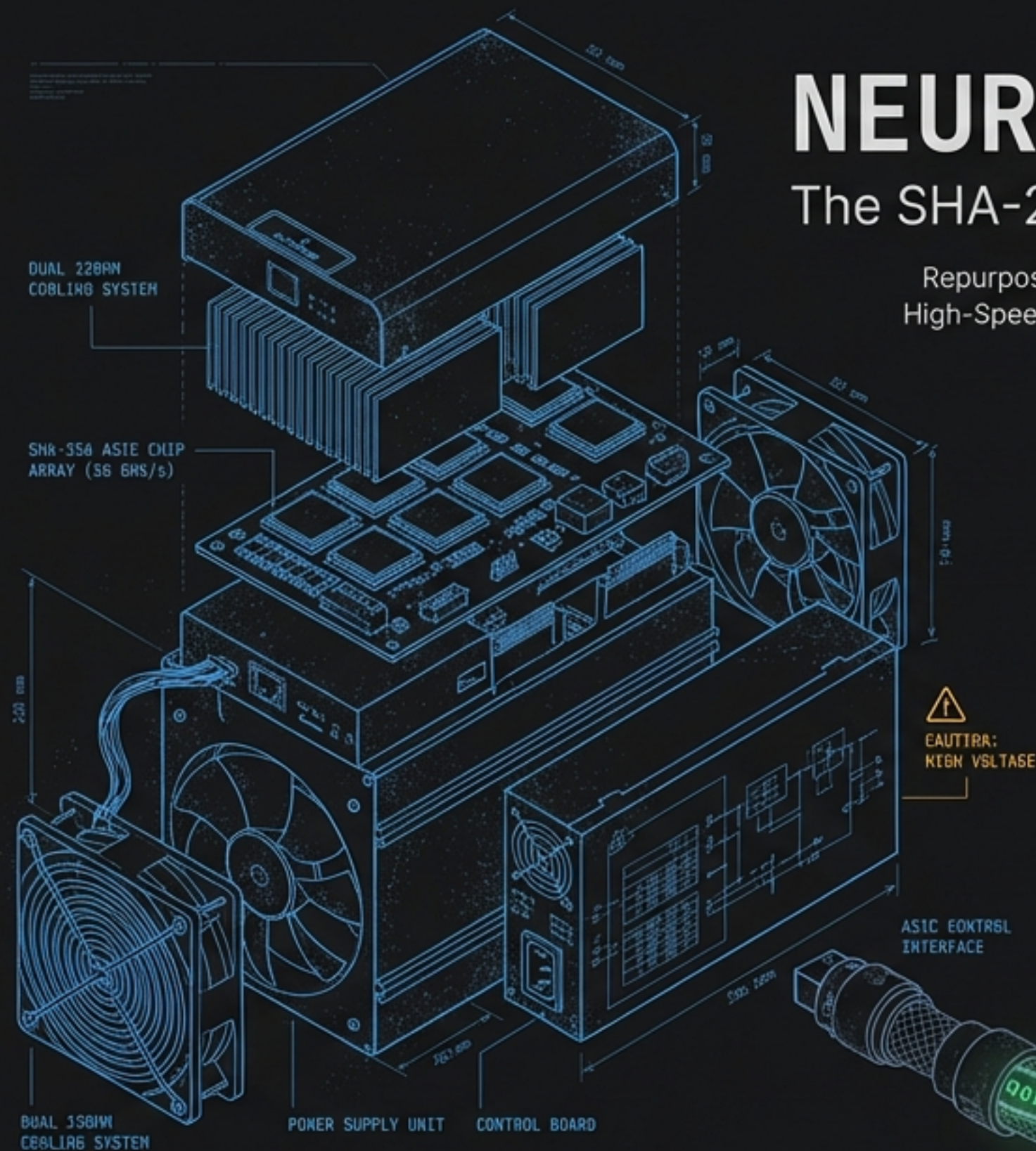


NEURAL MINER

The SHA-256 Inference Engine

Repurposing Antminer S3 E-Waste for High-Speed, Edge-Based LLM Inference.



Architecture v3.0 | The Neural eBPF Runtime

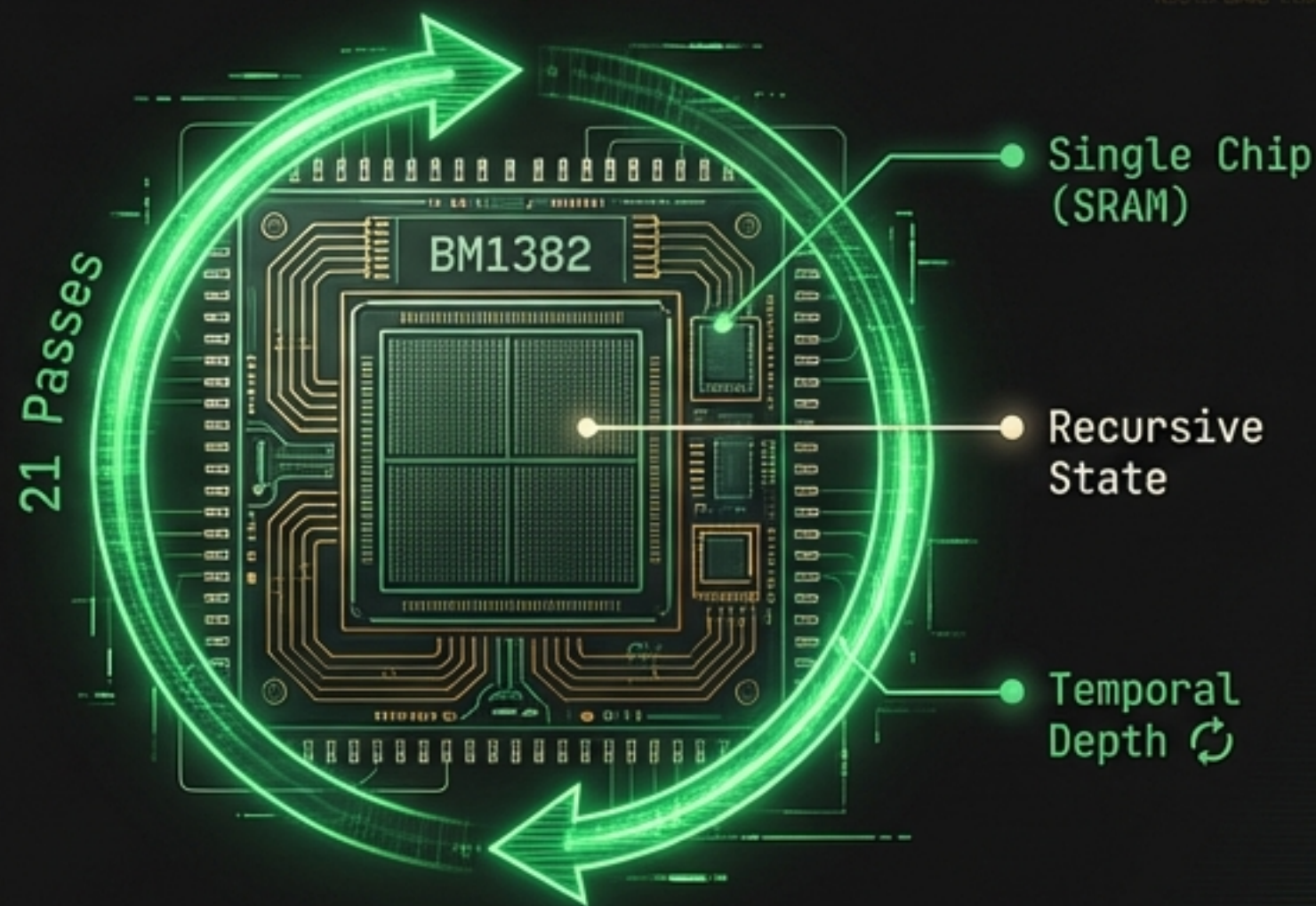
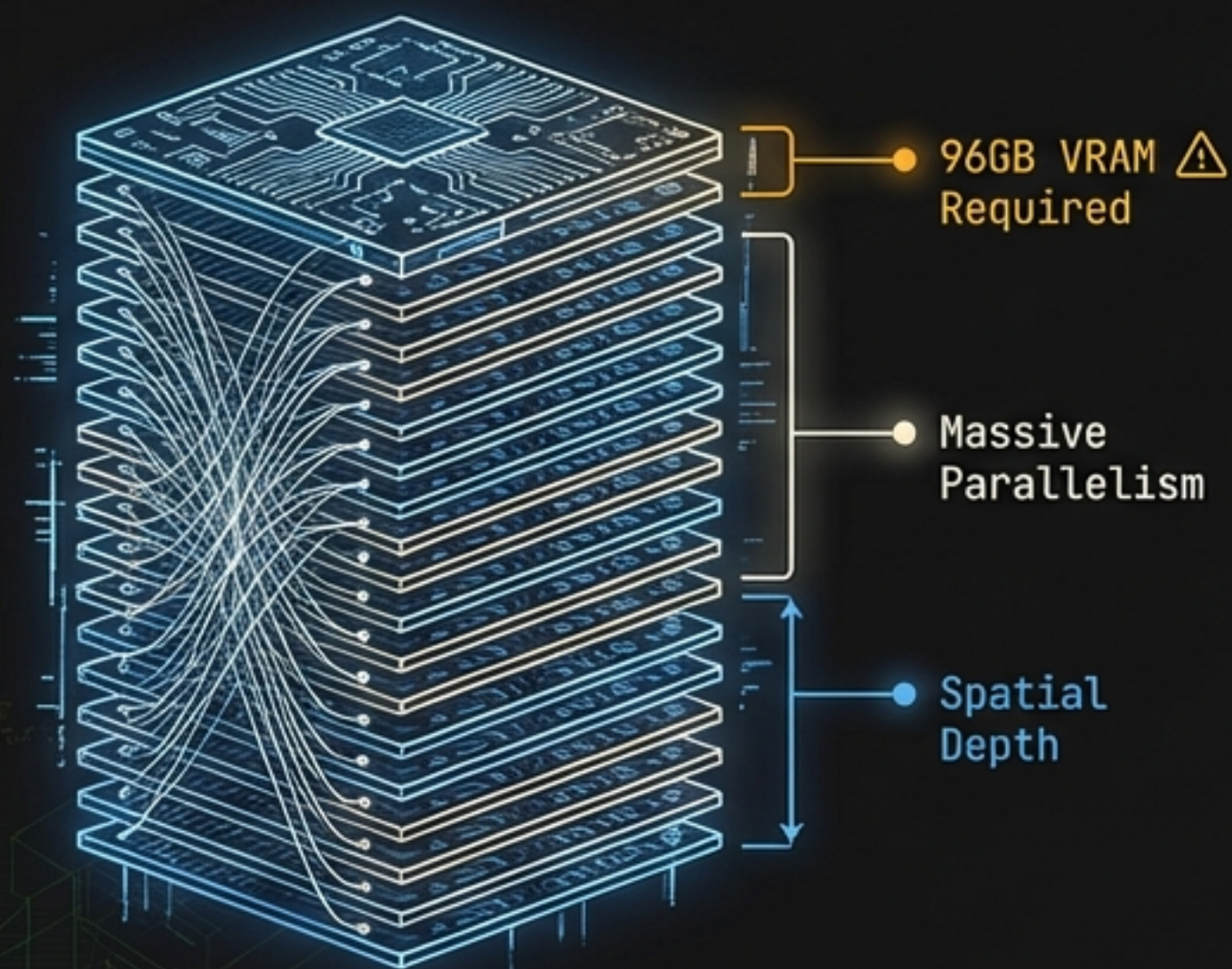
FROM HASHING TO REASONING

Replacing Spatial Depth with Temporal Depth

Standard Transformer (Spatial)

Efficiency Shift

Neural Miner
(Temporal)



Metric: Converting 500 GH/s hashrate into a 21-layer Recurrent Neural Network (RNN).

THE BM1382 TRAP

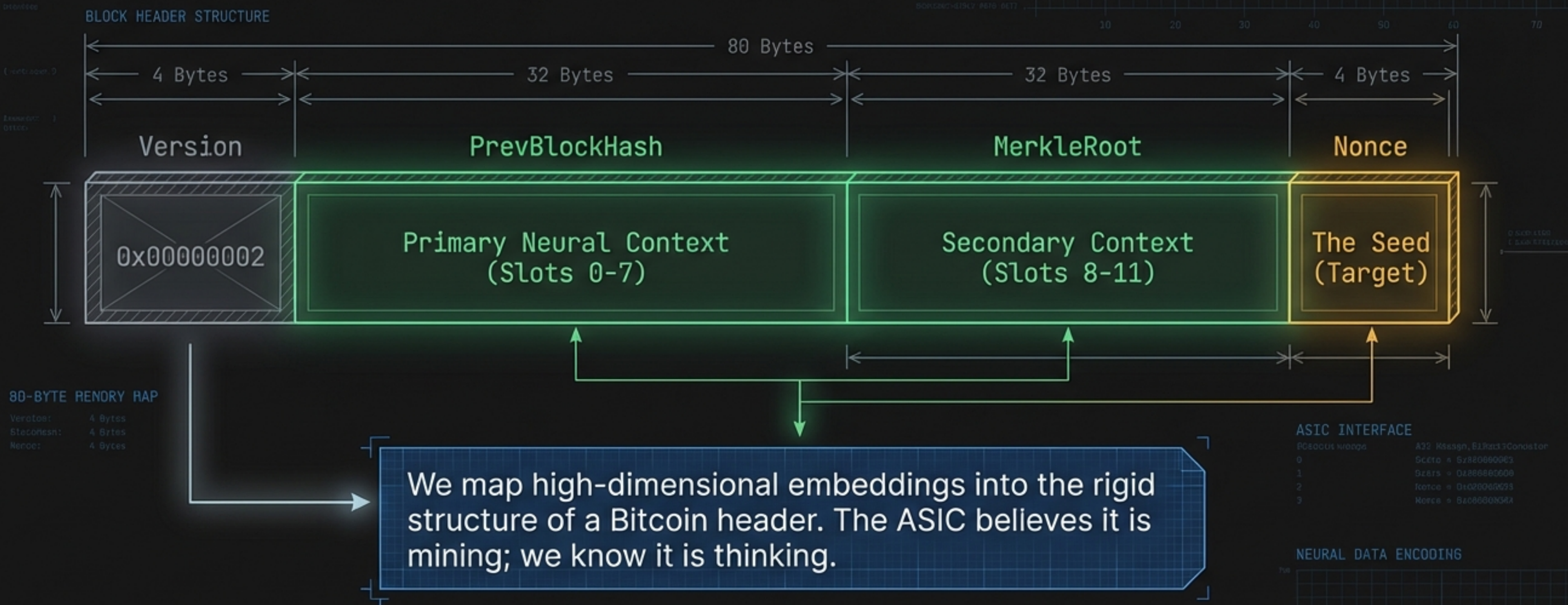
⚠ NO FLOATING POINT.
Cannot run FP16/FP32 math.

⚠ HARD-WIRED LOGIC.
Restricted to SHA256(SHA256(Header)).

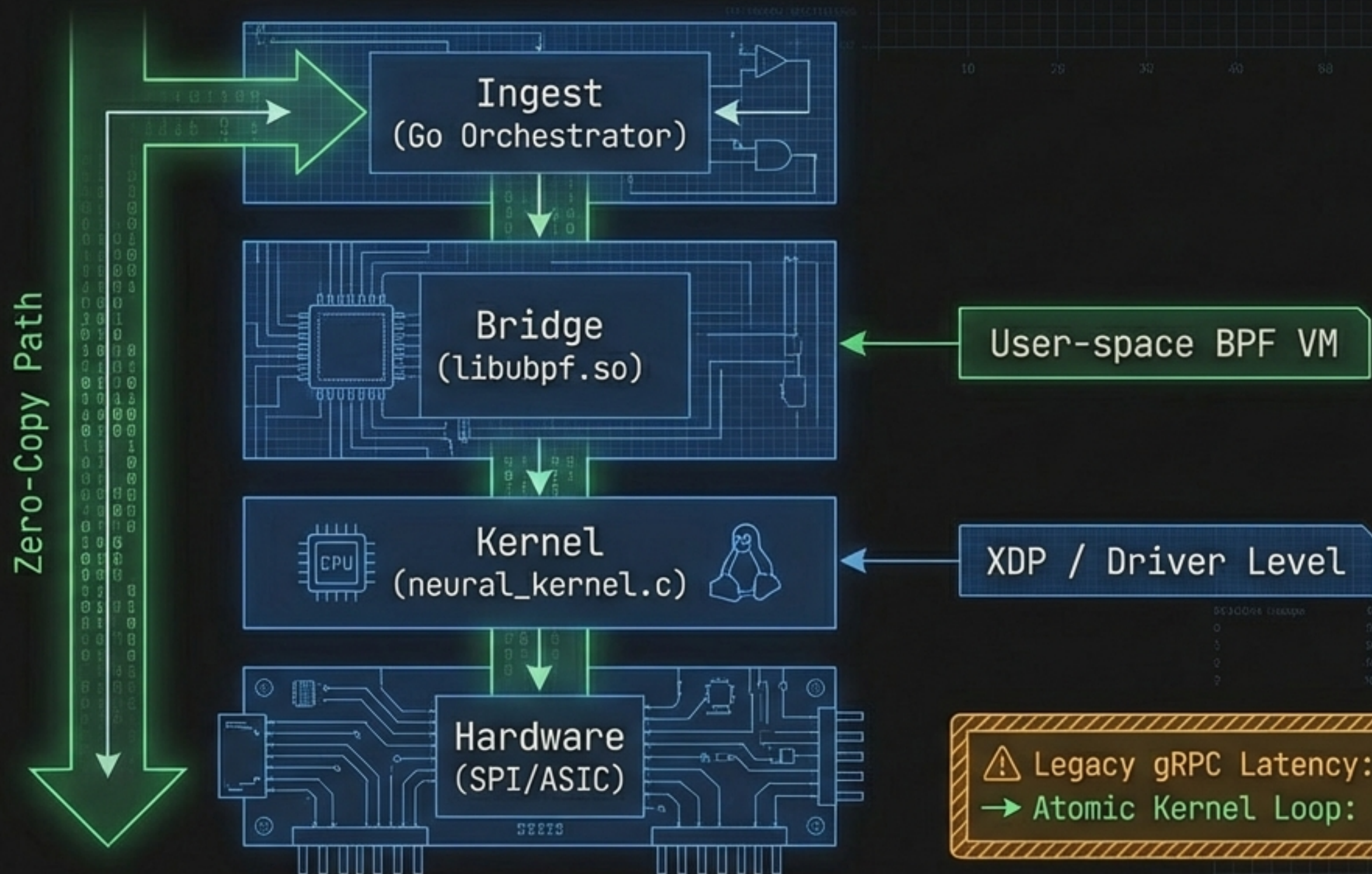
⚠ NO PYTHON RUNTIME.
Incompatible with
PyTorch/TensorFlow.

Implication: We cannot run the model. We must BECOME the mining job.

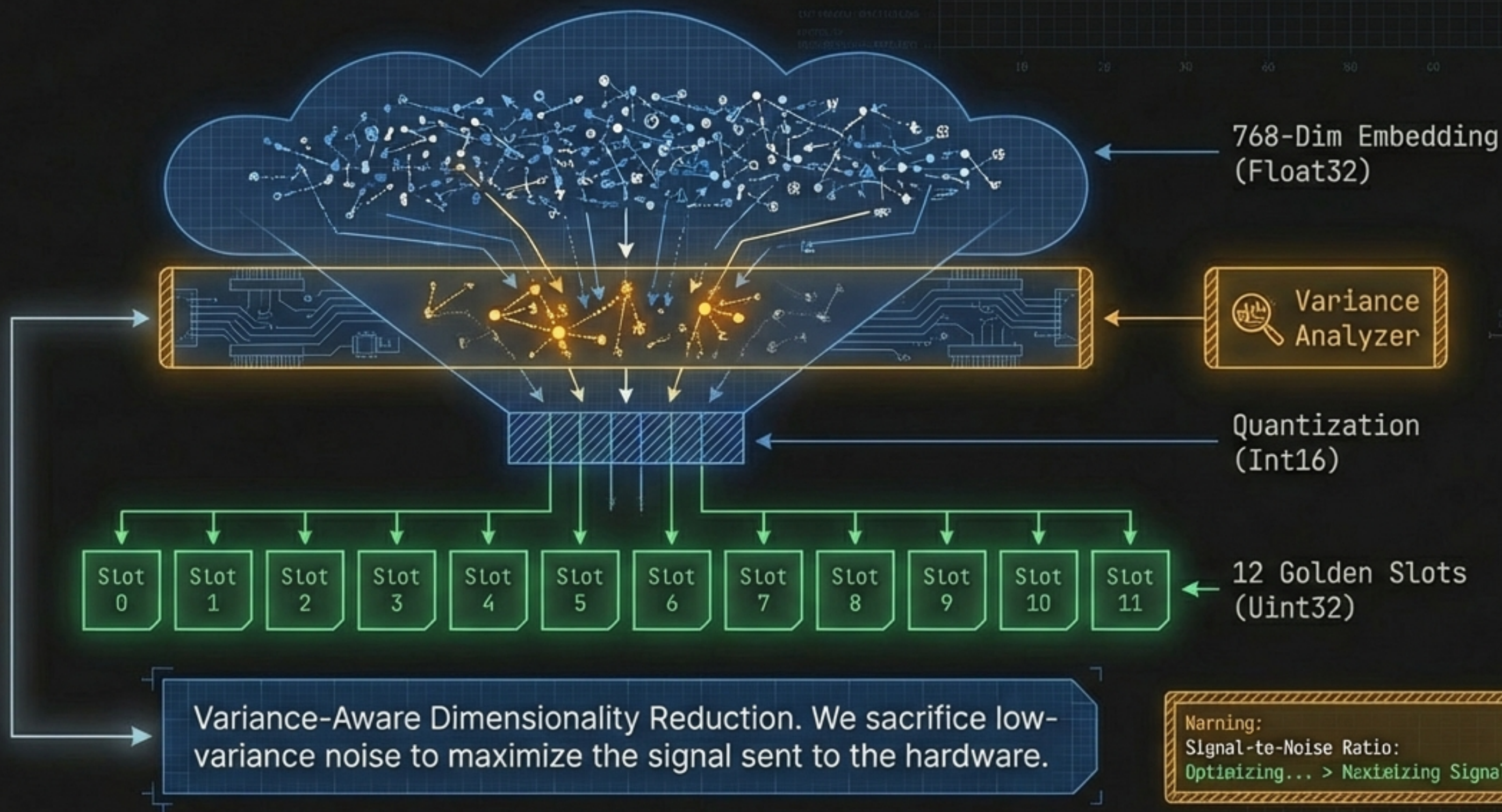
INNOVATION I: THE CAMOUFLAGE STRATEGY



INNOVATION II: THE NEURAL eBPF RUNTIME



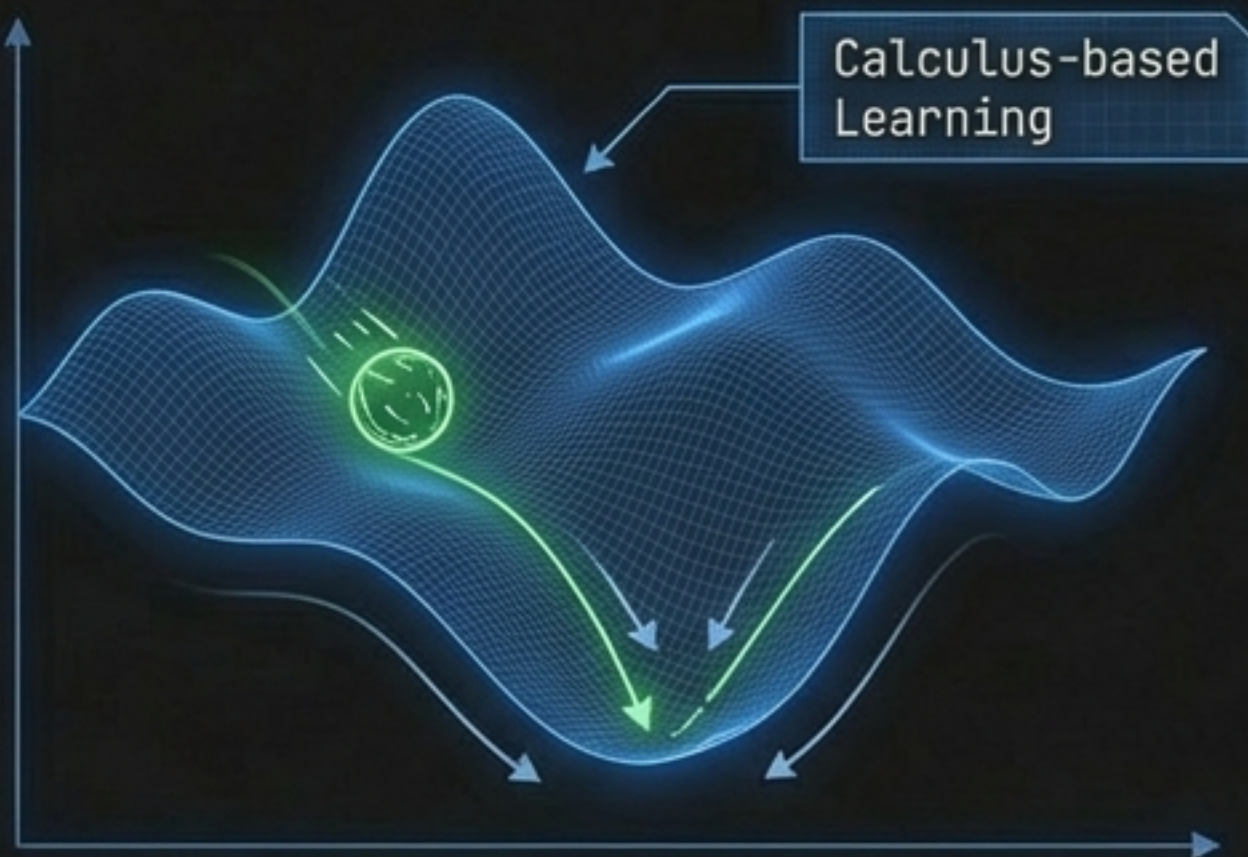
INNOVATION III: COMPRESSING MEANING



THE NON-DIFFERENTIABLE CLIFF

You Can't Backpropagate a Hash

Standard AI (Gradient Descent)



SHA-256 (Avalanche Effect)

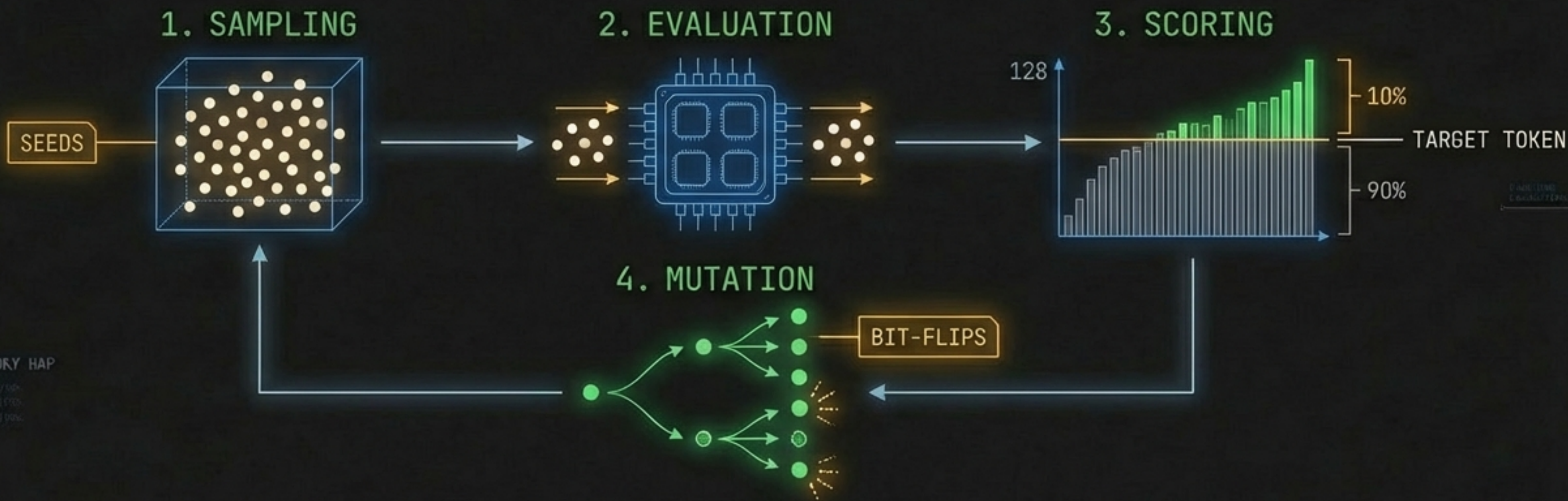


The Conflict: Flipping one bit changes the output randomly. There is no slope to follow.

Solution:
Abandon Backpropagation.
Embrace Evolution.

SOLUTION: EVO-GRPO HARNESS

Evolutionary Group Relative Policy Optimization

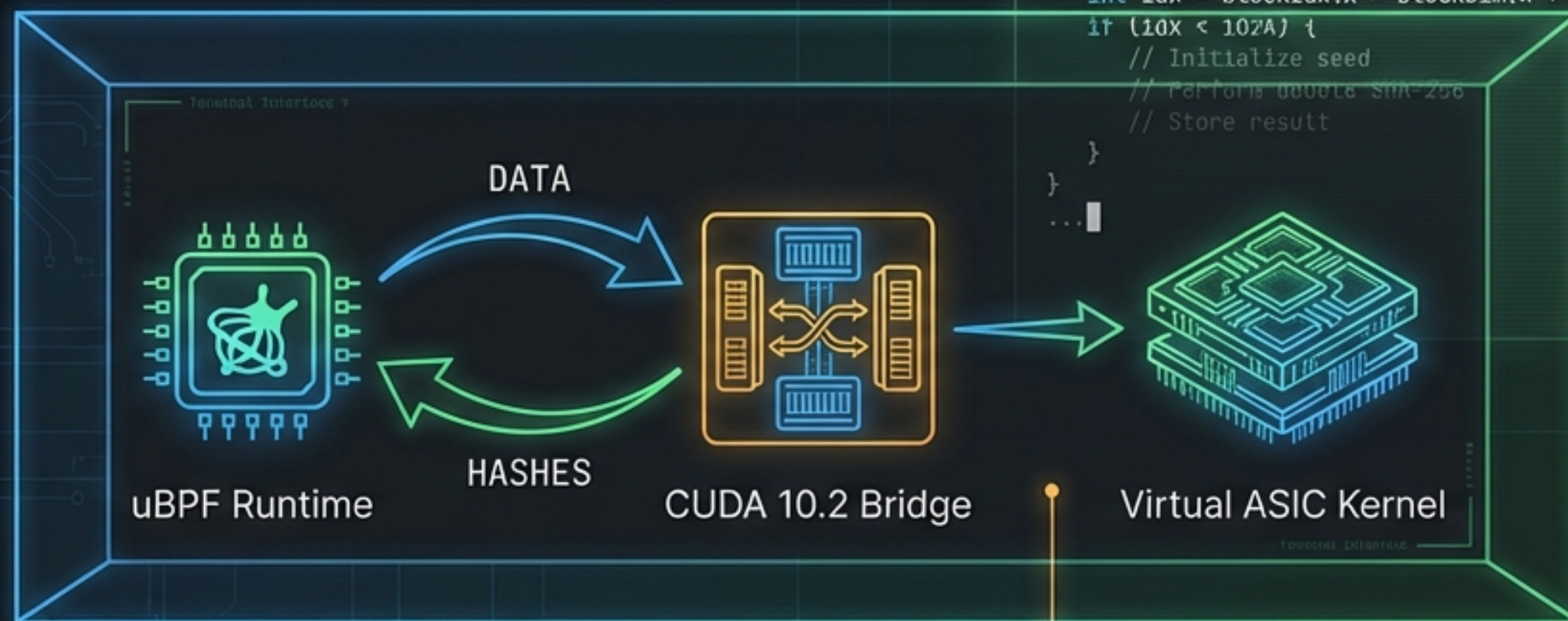


We replace the Critic Model with the Group Mean.
Survival of the fittest seed.

THE vHASHER SIMULATOR

The Digital Twin. Allows massive parallel training (1024 seeds) on standard GPUs before flashing to the physical ASIC.

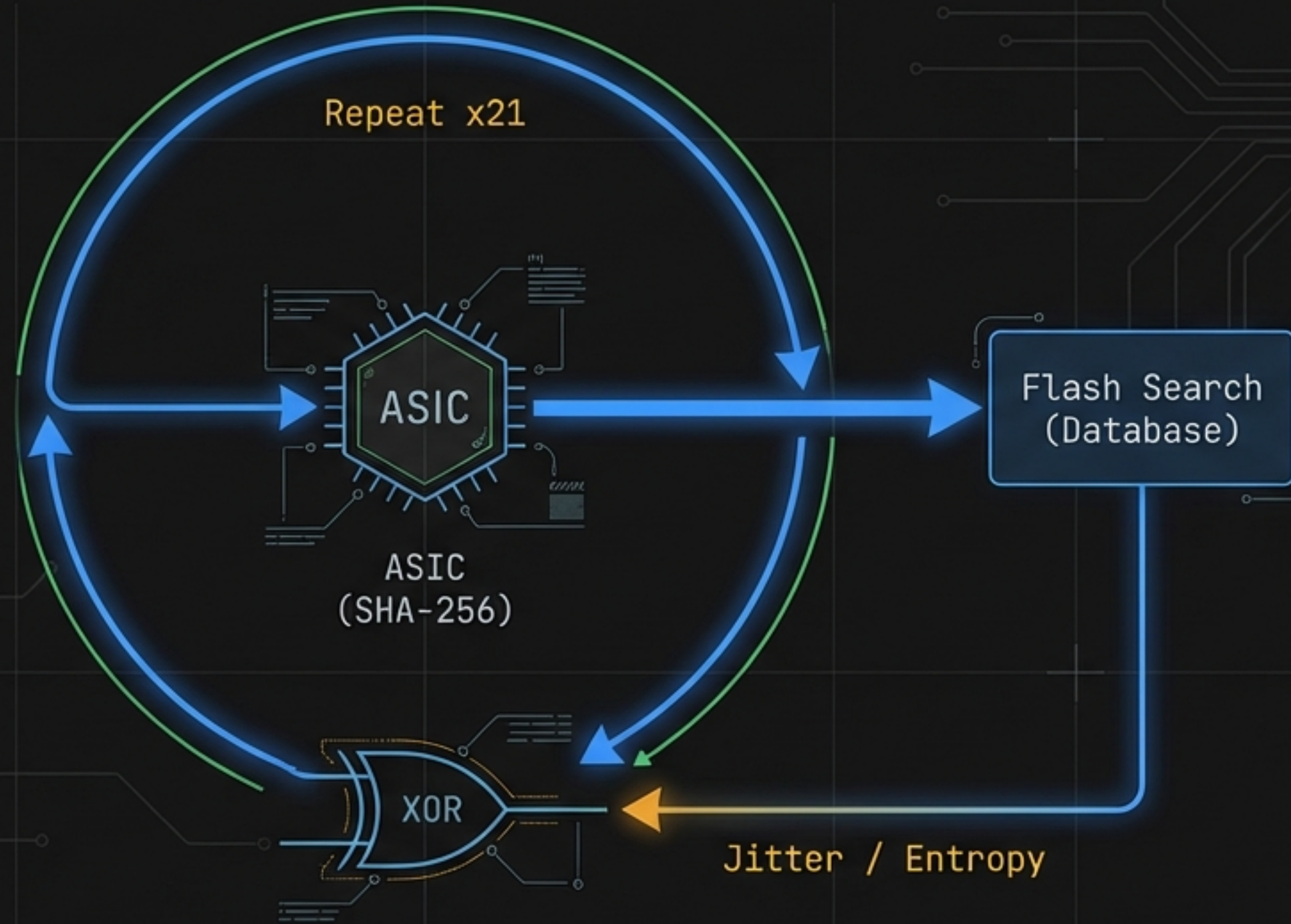
Docker Container



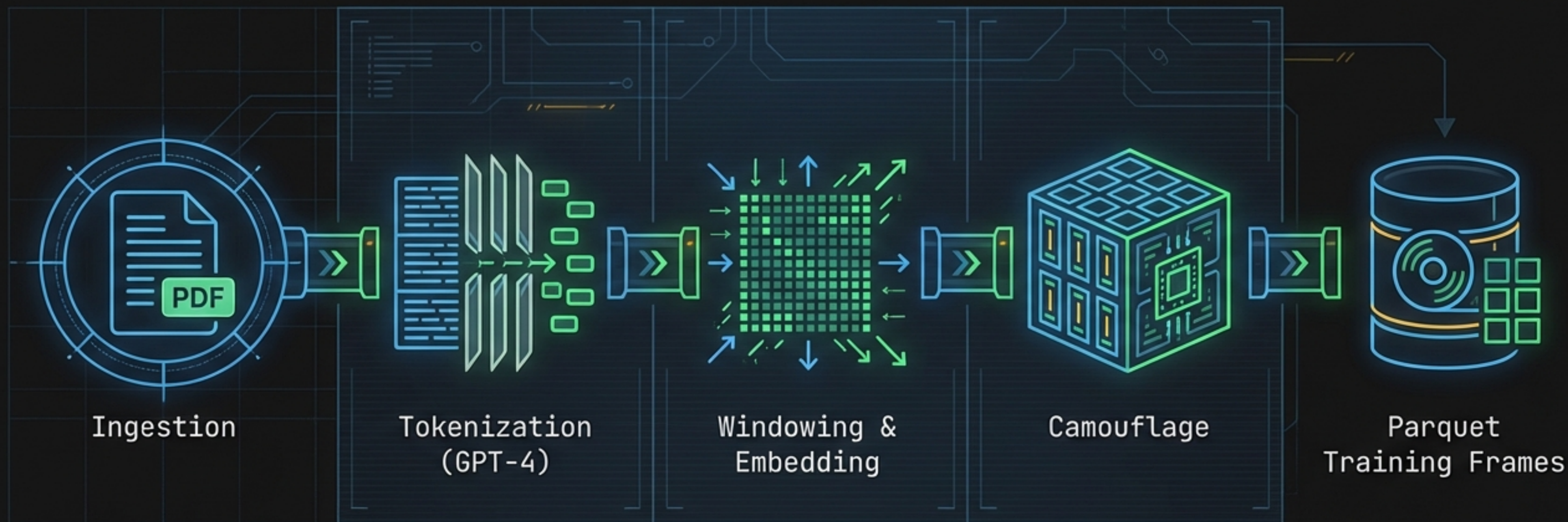
The Digital Twin. Allows massive parallel training (1024 seeds) on standard GPUs before flashing to the physical ASIC.

THE 21-PASS JITTER LOOP

Turns a static hash function into a dynamic, stateful reasoning engine.

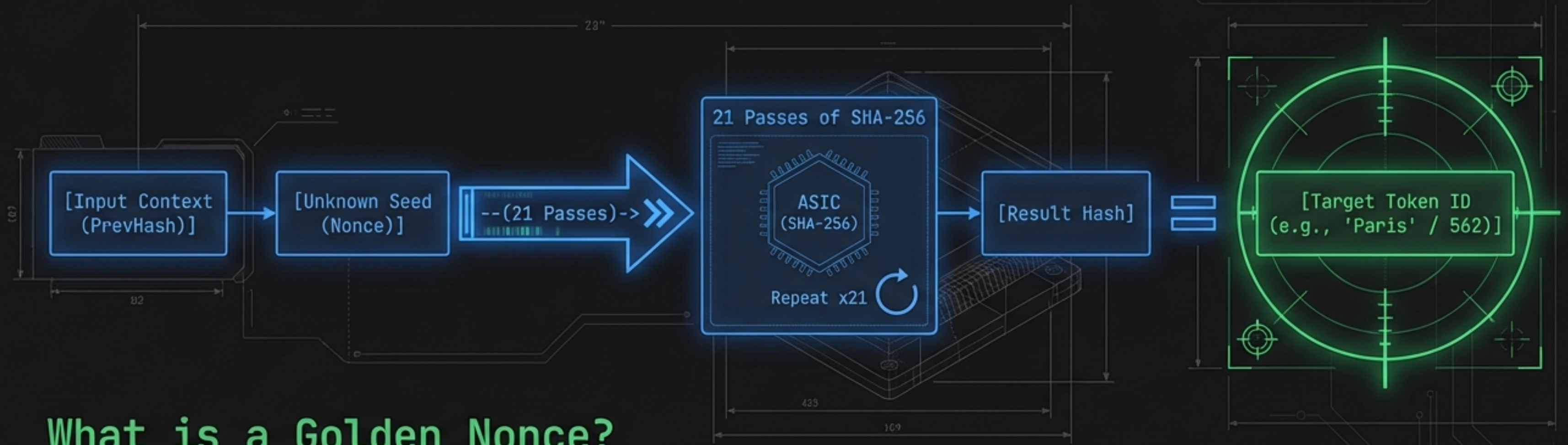


FROM PDF TO GOLDEN NONCE



The Data Structuring Engine tracks the transformation from raw text to hardware-ready training artifacts.

DEFINING THE BULLSEYE



What is a Golden Nonce?

A 32-bit seed that, after 21 passes of SHA-256, forces the ASIC output to match the Target Token ID.

Strategy: Use Prefix Matching (first 8-16 bits) to warm start training.

THE LEAN FIRMWARE STACK

⚠ Max Storage: 25MB



Portability:

The exact same binary runs on the GPU Simulator and the physical Antminer S3.

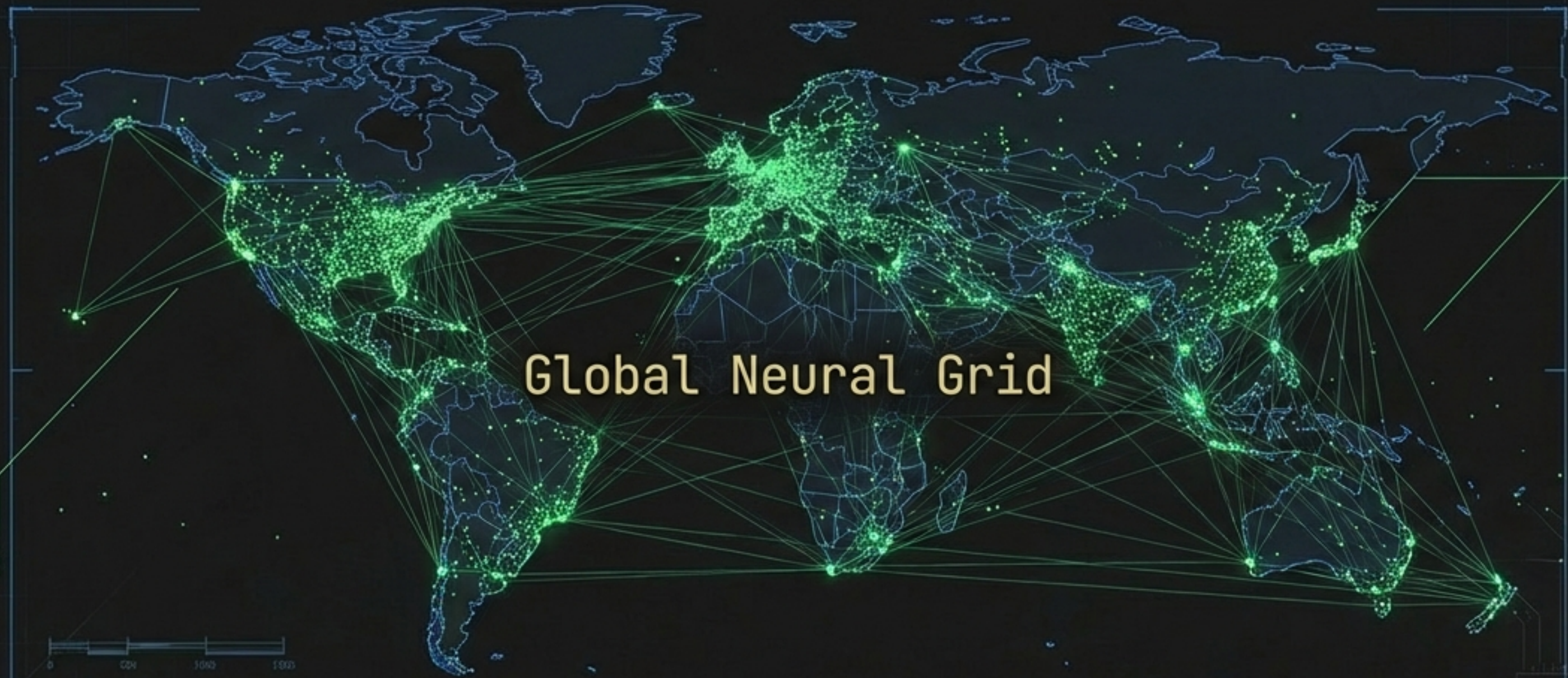
THE REALITY OF CHAOS

```
>  
[WARN] Failed to find winning seed for token 9827...  
[WARN] Failed to find winning seed for token 6706...  
[WARN] Search space exhaustion.  
>
```

The Optimization Fix

- ☐ Scale Population: 32 -> 512 seeds.
- ☐ Deep Time: 20 -> 200 generations.
- ☐ Soft Targets: Reward partial bit-matches (Hamming Distance).

THE FUTURE: DECENTRALIZED INTELLIGENCE



Optimized for E-Waste. Powered by Math. Secured by Proof-of-Work.

TURN TRASH INTO CHAT